Exact rank-one encoding and recovery for the Li–Jiang channel

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20 October 2026

Abstract

For the Li–Jiang noise family, we prove that the partial-trace decoder can be chosen in a jointly optimal rank-one encoder–decoder pair when the logical dimension is any $d \geq 2$, for every $0 < p < 1$, over all complex rank-one CPTNI encoders and all CPTNI decoders. The qutrit proof combines Li–Jiang's rank-one replacement bound with a sufficient condition for a unitary recovery to solve the full recovery SDP; a coarser fidelity estimate extends the argument to $d \geq 4$. In every logical dimension $d \geq 2$, we determine the full complex rank-one optimum with partial trace fixed, by a two-amplitude reduction and a specified quartic root. An explicit decoder SDP dual certificate proves that partial trace is an optimal decoder for every encoder in this real family. We give an exact algebraic certificate for the joint qutrit optimum at $p = 8/11$.

1 The channel and the joint question

Li and Jiang [1] construct a noise family whose unrestricted optimal fidelity requires higher-rank encoding. We study whether the partial-trace decoder can be chosen in a jointly optimal rank-one pair, and determine the exact fidelity when that decoder is fixed. Theorems 1 and 2 answer the joint question in every dimension. The argument uses a general recovery criterion and a coarse feasible fidelity above 3/4; it does not require a symmetry reduction of the nonconvex encoder set. Theorem 3 gives the fixed-decoder formula for all d, and Proposition 1 supplies a matching decoder certificate throughout the attaining family.

We use the model of Eqs. (18)–(23) of Ref. [1]. Let $L = S = \mathbb{C}^d$, $C = L \otimes S$, $\tau = I_d/d$, and

$$|\Omega\rangle = d^{-1/2} \sum_{i=0}^{d-1} |ii\rangle, \quad P = |\Omega\rangle\langle\Omega|, \quad D_i = I_d \otimes \langle i|.$$

For $0 < p < 1$ set

$$\begin{aligned} \lambda_1 &= 1 - \frac{p}{d^2-2}, \quad \lambda_4 = \frac{d^2-1+p}{d^2}, \quad \lambda_5 = \frac{1-p}{d^2}, \\ \lambda_2 &= \frac{(1-\lambda_1)(d^2\lambda_1 - 1)}{d^2\lambda_4}, \qquad \lambda_3 = \frac{2(1-\lambda_1)^2}{d^2\lambda_4}, \\ a &= \sqrt{\frac{d\lambda_4}{d^2-1}}, \quad b = \sqrt{d\lambda_5}, \quad Q = a(I-P) + bP. \end{aligned} \tag{1}$$

The noise action is

$$\mathcal{N}_p(X) = \lambda_3 \operatorname{tr}(X) I_C + (\lambda_1 - \lambda_3) \operatorname{tr}_S(QXQ) \otimes I_S + \lambda_2 \sum_{i,j} E_{ij} X E_{ij}^\dagger, \tag{2}$$

$$E_{ij} = \frac{|ij\rangle\langle\Omega| - \sqrt{\lambda_5} D_j^\dagger D_i Q}{\sqrt{\lambda_4}}.$$

A rank-one CPTNI encoder is $\mathcal{C}_V(X) = VXV^\dagger$, with $V \in \mathbb{C}^{d^2\times d}$ and $V^\dagger V \preceq I_d$. For logical Kraus operators L_k, fidelity means the unrenormalized quantity

$$F_{\mathrm{e}}(\tau, \Phi) = d^{-2} \sum_k |\operatorname{tr} L_k|^2. \tag{3}$$

Let $f_d(p)$ be the maximum with decoder tr_S fixed, and $F_1^{\mathrm{joint}}(p, d)$ the maximum allowing arbitrary CPTNI decoders as well.

2 The decoder reduction and the qubit case

Define

$$e_i = \frac{|i\rangle\langle\Omega| - \sqrt{\lambda_5} D_i Q}{\sqrt{\lambda_4}}.$$

Since $E_{ij} = (I \otimes |j\rangle) e_i$, summing over j in (2) gives the exact identity

$$\mathcal{N}_p(X) = \mathcal{T}_p(X) \otimes I_S/d, \qquad \mathcal{T}_p(X) = d\lambda_3 \operatorname{tr}(X) I_d + d(\lambda_1 - \lambda_3) \operatorname{tr}_S(QXQ) + d\lambda_2 \sum_i e_i X e_i^\dagger. \tag{4}$$

Thus every CPTP physical decoder $\mathcal{D} : C \to L$ induces the CPTP map $\mathcal{R}(X) = \mathcal{D}(X \otimes I_S/d)$, with $\mathcal{D}\mathcal{N}_p = \mathcal{R}\mathcal{T}_p$. Conversely $\mathcal{R}$ is realized by $\mathcal{D} = \mathcal{R} \operatorname{tr}_S$. The decoder Choi variable therefore has dimension $d^2 \times d^2$ rather than $d^3 \times d^3$.

Trace loss cannot help. Completing a CPTNI decoder to a CPTP decoder adds nonnegative terms to (3). For a fixed decoder, fidelity is a positive quadratic form $q(V)$. Complete the polar decomposition of a contraction as $V = UH$, $U^\dagger U = I_d$, $0 \preceq H \preceq I_d$. Then $V_\pm = U(H \pm i\sqrt{I - H^2})$ are isometries, and

$$\frac{q(V_+) + q(V_-)}{2} - q(V) = \frac{1}{4} q(V_+ - V_-) \geq 0.$$

An isometric encoder and CPTP decoder attain the joint maximum. This is the contraction argument of Lemma S1 of Ref. [1].

Theorem 1. *For every $0 < p < 1$, $F_1^{\mathrm{joint}}(p, 2) = f_2(p)$. In particular, the decoder can be chosen as tr_S in a globally optimal joint rank-one pair.*

Proof. For every qubit channel $\mathcal{A}$, a recovery maximizing its average input–output fidelity can be chosen unitary [2, Theorem 1]. Since $F_{\mathrm{av}}(\Phi) = (2F_{\mathrm{e}}(I_2/2, \Phi) + 1)/3$ for a qubit CPTP map, the same is true for our objective. This theorem applies to $\mathcal{A} = \mathcal{T}_p \mathcal{C}_V$ for each isometry V. If M_k are Kraus operators of $\mathcal{T}_p$ and U is such a unitary recovery, cyclicity gives

$$\frac{1}{4} \sum_k |\operatorname{tr}(UM_kV)|^2 = \frac{1}{4} \sum_k |\operatorname{tr}(M_kVU)|^2.$$

The matrix VU is still an isometry. The unitary recovery is therefore absorbed into the encoder, proving the equality. The reverse inequality follows from feasibility of tr_S. □

The cited qubit theorem follows by removing the recovery's Bloch translation without changing its objective and then using that every unital qubit channel is a mixture of unitary channels [2, Appendix A]. Its conclusion does not extend to arbitrary channels in dimension three.

2.1 A recovery criterion and joint optimality in every dimension

For a CPTP channel $\mathcal{A} : \mathsf{L}(\mathbb{C}^d) \to \mathsf{L}(\mathbb{C}^d)$, write

$$u(\mathcal{A}) = \max_{U \text{ unitary}} F_{\mathrm{e}}(I_d/d, \mathrm{Ad}_U \mathcal{A}), \qquad t(\mathcal{A}) = \max_{\mathcal{R} \text{ CPTP}} F_{\mathrm{e}}(I_d/d, \mathcal{R}\mathcal{A}).$$

The unitary quasi-inverse theorem used above is special to qubits: higher-dimensional channels can require nonunitary quasi-inverses [3]. The following sufficient condition applies in any dimension.

Lemma 1. *If $f = u(\mathcal{A}) > 0$ and*

$$(d+1)f - (d-1)\sqrt{f} - 1 \geq 0, \tag{5}$$

then $t(\mathcal{A}) = u(\mathcal{A})$. If $d = 3$ and $t(\mathcal{A}) \geq 3/4$, an optimal recovery can be chosen unitary.

Proof. Absorb a maximizing unitary into $\mathcal{A}$, so identity recovery maximizes the unitary objective. For Kraus operators A_i, stationarity under $\exp(i\theta H)$ for every Hermitian H implies that

$$B = \sum_i \overline{\operatorname{tr} A_i}\, A_i \quad \text{is Hermitian.}$$

Indeed the derivative of the fidelity is $-2d^{-2} \operatorname{Im} \operatorname{tr}(HB)$. The operator $A_0 = B/(d\sqrt{f})$ is a normalized linear combination of the Kraus operators, hence can be included in a Kraus representation and is a contraction. It is Hermitian and has trace $d\sqrt{f}$. Since each of its eigenvalues is at most one,

$$\lambda_{\min}(A_0) \geq d\sqrt{f} - (d-1). \tag{6}$$

Use input-first vectorization $|M\rangle\!\rangle = \sum_j |j\rangle \otimes M|j\rangle$. The recovery SDP has objective matrix $K = J(\mathcal{A}^\dagger)/d^2$, as in (22). Set

$$Y = \frac{\sqrt{f}}{d} A_0^T, \qquad S = Y \otimes I - K, \qquad e = |I\rangle\!\rangle/\sqrt{d}, \quad P_\perp = I - ee^\dagger.$$

Then $\operatorname{tr} Y = f$ and $Se = 0$, since $K|I\rangle\!\rangle = d^{-2}|B^\dagger\rangle\!\rangle$ and $(Y \otimes I)|I\rangle\!\rangle = (\sqrt{f}/d)|A_0\rangle\!\rangle$. Trace preservation gives $\operatorname{tr} K = 1/d$, and $e^\dagger K e = f/d$, so $\operatorname{tr}(P_\perp K P_\perp) = (1-f)/d$. For every unit vector $v \perp e$, positivity of K and (6) give

$$v^\dagger S v \geq \lambda_{\min}(Y) - \operatorname{tr}(P_\perp K P_\perp) \geq \frac{(d+1)f - (d-1)\sqrt{f} - 1}{d}.$$

Thus $S \succeq 0$ under (5); Y and identity recovery are matching dual and primal solutions.

For the qutrit assertion, take a recovery attaining $t = t(\mathcal{A})$. Cyclicity of Kraus traces gives $F_{\mathrm{e}}(I_3/3, \mathcal{R}\mathcal{A}) = F_{\mathrm{e}}(I_3/3, \mathcal{A}\mathcal{R})$. Apply the CPTP version of Li–Jiang's Theorem 1 [1] with channel $\mathcal{A}$ and encoder $\mathcal{R}$, both acting on $\mathbb{C}^3$. Its rank-one CPTP replacement is a square isometry, hence a unitary. With $\varepsilon = 1 - t \leq 1/4$ the bound is

$$f \geq t - \frac{2}{9}\bigl(1 - \sqrt{1 - 3\varepsilon}\bigr)^2 \geq \frac{3}{4} - \frac{1}{18} = \frac{25}{36}.$$

For $\sqrt{f} \geq 5/6$, $4f - 2\sqrt{f} - 1 \geq 1/9 > 0$. The first assertion therefore applies. □

The same replacement theorem also gives, in every dimension,

$$u(\mathcal{A}) \geq t(\mathcal{A})^2. \tag{7}$$

For its first branch this follows from $\ell_{1/d}(\varepsilon) \leq \varepsilon$ and the lower bound $(1-\ell_{1/d}(\varepsilon))^2$ in its proof, Eq. (8); its second branch is Eq. (11). Consequently, if $t(\mathcal{A}) \geq (d+2)/(d+3)$, condition (5) holds: putting $r = (d+2)/(d+3)$ gives

$$(d+1)r^2 - (d-1)r - 1 = \frac{d+1}{(d+3)^2} > 0,$$

and the quadratic $(d+1)s^2 - (d-1)s - 1$ increases for $s \geq r$.

Theorem 2. *For every $d \geq 2$ and $0 < p < 1$, $F_1^{\text{joint}}(p,d) = f_d(p)$. An optimal rank-one pair consists of the complex-space fixed-decoder optimizer of Theorem 3 and tr_S.*

Proof. The case $d = 2$ is Theorem 1. First take $d = 3$. The product isometry $V_0|j\rangle = |j0\rangle$, followed by tr_S, has fidelity

$$F_0(q) = \frac{-128q^4 + 256q^3 + 711q^2 + 752q + 3512}{567(q^2+8)}, \qquad q = \sqrt{\frac{8(1-p)}{8+p}} \in (0,1). \tag{8}$$

This follows by substitution in (2), or in the full encoder quadratic form (17). Its exact positive gap is

$$F_0(q) - \frac{3}{4} = \frac{440 + 3008q + 1143q^2 + 512q^3 + 512q^3(1-q)}{2268(q^2+8)} > 0. \tag{9}$$

Take a jointly maximizing isometry V and CPTP reduced recovery, whose existence follows from compactness and the reductions above. The square qutrit channel $\mathcal{A} = \mathcal{T}_p\mathcal{C}_V$ satisfies $t(\mathcal{A}) \geq F_0(q) > 3/4$. Lemma 1 therefore supplies a unitary optimal recovery U. Cyclicity gives

$$\frac{1}{9}\sum_k |\,\mathrm{tr}(UM_kV)|^2 = \frac{1}{9}\sum_k |\,\mathrm{tr}(M_kVU)|^2 \leq f_3(p),$$

where M_k are Kraus operators of $\mathcal{T}_p$, and VU is still an isometry. The reverse inequality follows from feasibility of tr_S.

For $d \geq 4$, the same product isometry has fidelity

$$F_0 = \lambda_3 + dA - \frac{d^2-1}{d^2}C_0,$$

where A, C_0 are defined in (11). Using $0 < C_0 < 2A/d$, $A = 1/d - d\lambda_3$, and $\lambda_3 = 2p^2/[(d^2-2)^2(d^2-1+p)] \leq 2/[d^2(d^2-2)^2]$, we obtain

$$F_0 \geq m_d := \frac{d^6 - 4d^4 + 4d^2 - 2}{d^4(d^2-2)} > \frac{d+2}{d+3}. \tag{10}$$

For completeness, the numerator of the last difference, over the positive denominator $d^4(d+3)(d^2-2)$, is

$$d^6 - 2d^5 - 8d^4 + 4d^3 + 12d^2 - 2d - 6.$$

Putting $k = d - 4 \geq 0$ rewrites this numerator as

$$k^6 + 22k^5 + 192k^4 + 836k^3 + 1852k^2 + 1822k + 434 > 0.$$

The bound on λ_3 follows since $p^2/(d^2-1+p)$ increases on $[0,1]$. Every jointly maximizing isometry thus produces a square channel with $t(\mathcal{A}) \geq F_0 > (d+2)/(d+3)$. Equations (7) and (5) supply a unitary optimal recovery, and the same trace cyclicity absorbs it into the encoder. □

3 The exact fixed-decoder optimum in every dimension

Put $n = d - 1$ and define

$$A = \frac{1}{d} - d\lambda_3, \qquad C_0 = \frac{2A}{d} - 2(\lambda_1 - \lambda_2 - \lambda_3)\sqrt{\frac{\lambda_4\lambda_5}{d^2-1}}. \tag{11}$$

These satisfy $A > 0$ and $0 < C_0 < 2A/d$. For example, with

$$q = \frac{b}{a} = \sqrt{\frac{(d^2-1)(1-p)}{d^2-1+p}}, \qquad p = \frac{(d^2-1)(1-q^2)}{d^2-1+q^2}, \tag{12}$$

direct simplification gives

$$A = \frac{(d^2-1-q^2)[d^4-4d^2+2+2(d^2-1)q^2]}{d(d^2-2)^2(d^2+q^2-1)}, \qquad \frac{C_0}{2A/d} = \frac{(1-q)(d^2+q-1)}{d^2-1-q^2} \in (0,1).$$

Theorem 3. *For all $d \geq 2$ and $0 < p < 1$, $f_d(p)$ is attained by*

$$V|0\rangle = x|00\rangle + z\sum_{i=1}^{n}|ii\rangle, \qquad V|j\rangle = |j0\rangle \ (j > 0), \qquad x \geq 0,\ z \leq 0,\ x^2 + nz^2 = 1. \tag{13}$$

Set

$$\delta = C_0(A/d - C_0/4), \quad R = A^2 - AC_0 + dC_0^2/4, \quad D(\mu) = \mu^2 - A\mu + n\delta.$$

There is a unique root $A < \mu < dA$ of

$$D(\mu)^2 - n^2(R\mu^2 - 2nA\delta\mu + nd\delta^2) = 0. \tag{14}$$

For that root,

$$f_d(p) = \lambda_3 + \frac{1}{d}\left(An^2 + \mu + \frac{n^2(R\mu - nA\delta)}{D(\mu)}\right), \tag{15}$$

$$x_* = \frac{n[(A - C_0/2)\mu - n\delta]}{D(\mu)}, \qquad z_* = -\frac{nC_0[\mu - (2A/d - C_0/2)]}{2D(\mu)}. \tag{16}$$

This is also the joint optimum of Theorem 2.

3.1 Reduction over the full complex encoder space

For arbitrary complex V, write $r_i = \mathrm{tr}(D_i V)$ and $z_i = \langle\Omega|V|i\rangle$. Expanding the three Kraus groups of (2) yields

$$F(V, \mathrm{tr}_S) = \frac{\lambda_3}{d}\|V\|_F^2 + \frac{1}{d}\left[A\|r\|^2 - \sqrt{d}\, C_0\, \mathrm{Re}(r^\dagger z) + C_0\|z\|^2\right]. \tag{17}$$

One way to check the expansion is to use $C_i = QD_i^\dagger$ and $H_i = (|\Omega\rangle\langle i| - \sqrt{\lambda_5}C_i)/\sqrt{\lambda_4}$. The two nonconstant Kraus contributions are $(\lambda_1 - \lambda_3)d^{-1}\sum_i |\langle C_i, V\rangle|^2$ and $\lambda_2 d^{-1}\sum_i |\langle H_i, V\rangle|^2$. Inserting $Q = aI + (b-a)P$ gives (17).

The first term is λ_3 on isometries. Write the bracket as $q(V)$. Its 2×2 coefficient matrix is

$$G = \begin{pmatrix} A & -\sqrt{d}C_0/2 \\ -\sqrt{d}C_0/2 & C_0 \end{pmatrix} \succ 0.$$

Duality for this positive quadratic form and the operator/nuclear norm duality give

$$\max_{V^\dagger V=I} q(V) = \max_{(\alpha,\beta)\neq 0} \frac{\|I\otimes|\alpha\rangle + |\Omega\rangle\beta^T\|_*^2}{(\alpha,\beta)^\dagger (G^{-1}\otimes I_d)(\alpha,\beta)}. \tag{18}$$

Indeed the inner linear functional is $\alpha^\dagger r + \beta^\dagger z$; its maximum absolute value over isometries is the nuclear norm of the displayed rectangular matrix, attained by its polar isometry. Maxima interchange after normalizing the positive denominator.

Let $a_0 = \|\alpha\|$, $b_0 = \|\beta\|$, $c_0 = \mathrm{Re}(\alpha^\dagger\beta)$, and $k = |\alpha^\dagger\beta|^2$. The denominator in (18) depends only on a_0, b_0, c_0. The Gram matrix of its numerator is

$$a_0^2 I + \overline{\beta}\beta^T + \frac{\overline{\alpha}\beta^T + \overline{\beta}\alpha^T}{\sqrt{d}}.$$

At most two eigenvalues differ from a_0^2. Their sum and product are

$$T = 2a_0^2 + b_0^2 + 2c_0/\sqrt{d}, \qquad P_2 = a_0^4 + a_0^2 b_0^2(1-1/d) + 2a_0^2 c_0/\sqrt{d} + k/d.$$

Consequently the nuclear norm is $(d-2)a_0 + \sqrt{T + 2\sqrt{P_2}}$, which increases with k at fixed a_0, b_0, c_0. Collinear complex vectors with a suitable relative phase realize $k = a_0^2 b_0^2$ and preserve these other three invariants. Zero vectors are included by continuity. Under $M \mapsto (u\otimes\overline{u})Mu^\dagger$, both vectors transform by $\overline{u}$, so the collinear vectors can be placed along $|0\rangle$. The polar isometry then has exactly the supports in (13), with complex amplitudes x, z and a common amplitude z on $|ii\rangle$, $i > 0$. Vanishing singular values admit extensions on these same supports. Thus the complex family attains the full maximum without averaging encoders.

Substitution in (17) gives

$$\begin{aligned} q(V) = {} & An^2 + (A - nC_0/d)|x|^2 + (n^2 C_0/d)|z|^2 \\ & - \frac{n(d-2)C_0}{d}\,\mathrm{Re}(\overline{x}z) + n(2A - C_0)\,\mathrm{Re}\,x - n^2 C_0\,\mathrm{Re}\,z. \end{aligned} \tag{19}$$

Replacing x, z by $|x|, -|z|$ increases or preserves every phase-dependent term. This proves the real reduction in (13).

3.2 A two-variable maximization and its exact dual

Let $w = (x, z)^T$, $W = \mathrm{diag}(1, n)$ and

$$H = \begin{pmatrix} A - nC_0/d & -n(d-2)C_0/(2d) \\ -n(d-2)C_0/(2d) & n^2 C_0/d \end{pmatrix}, \qquad h = \begin{pmatrix} n(A - C_0/2) \\ -n^2 C_0/2 \end{pmatrix}.$$

The objective is $\lambda_3 + d^{-1}(An^2 + w^T H w + 2h^T w)$ on $w^T W w = 1$. The positive matrix $B = W^{-1/2} H W^{-1/2}$ has trace A and determinant $n\delta > 0$, hence $\lambda_{\max}(B) < A$. The secular function

$$\|(\mu I - B)^{-1} W^{-1/2} h\|^2$$

strictly decreases for $\mu > A$. At $\mu = A$, its first coordinate equals $x(A) = A(A - C_0/2)/\delta - n > 1$. The last inequality follows from

$$A(A - C_0/2) - d\delta = A^2 - 3AC_0/2 + dC_0^2/4 > 0 \quad (0 < C_0 < 2A/d,\ d \geq 2).$$

At $\mu = dA$, its norm is less than one because $\|W^{-1/2}h\| = n\sqrt{R} < nA$ and $\lambda_{\max}(B) < A$. There is therefore a unique secular root in (A, dA). Solving $(\mu W - H)w = h$ gives (16); imposing $w^T W w = 1$ gives (14), with no extraneous roots in that interval. The formulas give $x_* > 0$, $z_* < 0$.

For a directly checkable certificate, set

$$S = \mu W - H \succ 0, \quad w_* = S^{-1}h, \quad \eta = h^T S^{-1} h = \frac{n^2(R\mu - nA\delta)}{D(\mu)}, \quad Z = \begin{pmatrix} \eta & -h^T \\ -h & S \end{pmatrix} \succeq 0. \tag{20}$$

The Schur complement is zero, and $Z(1, w_*^T)^T = 0$. On $w^T W w = 1$ the exact identity

$$f_d(p) - F(V(w), \mathrm{tr}_S) = d^{-1}(w - w_*)^T S (w - w_*) \geq 0 \tag{21}$$

proves the bound with equality at w_*. Equivalently Z is dual to the lifted quadratic program and $(1, w_*^T)^T(1, w_*^T)$ is a matching rank-one primal matrix. This completes Theorem 3.

4 A decoder certificate for every encoder in the family

For a fixed isometry V, put $\mathcal{A} = \mathcal{T}_p \mathcal{C}_V$ and $K = J(\mathcal{A}^\dagger)/d^2$, where the input-first Choi convention is $J(\mathcal{R}) = \sum_{ij} |i\rangle\langle j| \otimes \mathcal{R}(|i\rangle\langle j|)$. The decoder SDP and its dual are

$$\max_{X \succeq 0,\ \mathrm{tr}_{\mathrm{out}} X = I_d} \mathrm{tr}(KX), \qquad \min_{Y = Y^\dagger,\ Y \otimes I_d \succeq K} \mathrm{tr}\, Y. \tag{22}$$

These normalizations follow immediately from (3).

Proposition 1. *For every real encoder* (13)*, the identity reduced recovery, and therefore the physical decoder* tr_S*, is optimal among all decoders.*

Proof. Write $z = -t$, $t \geq 0$, $s = x - nt$, and put

$$\alpha = d\lambda_3, \quad \beta = dA, \quad c = d^2 C_0/2 \in (0, \beta),$$

$$a_0 = \alpha + \beta x^2 - c(2sx/d - 2s^2/d^2), \quad b_0 = \alpha + \beta t^2 + c(2st/d + 2s^2/d^2), \quad g = \beta x - cs/d.$$

With $P_0 = |0\rangle\langle 0|$, $\Pi = I - P_0$, direct evaluation of (4) gives

$$\mathcal{A}(|0\rangle\langle 0|) = a_0 P_0 + b_0 \Pi, \quad \mathcal{A}(|j\rangle\langle j|) = \alpha I + \beta |j\rangle\langle j| \ (j > 0),$$

$$\mathcal{A}(|0\rangle\langle j|) = g|0\rangle\langle j| \ (j > 0), \qquad \mathcal{A}(|i\rangle\langle j|) = \beta |i\rangle\langle j| \ (i, j > 0,\ i \neq j).$$

The adjoint relations determine the remaining matrix units. Take

$$Y = d^{-2}\big[(a_0 + ng)P_0 + (g + n\beta + \alpha)\Pi\big]. \tag{23}$$

On the span of $|00\rangle$ and $n^{-1/2}\sum_{j>0}|jj\rangle$, and on its orthogonal invariant subspaces, the slack is

$$\begin{aligned} d^2(Y \otimes I - K) \ \cong\ & g \begin{pmatrix} n & -\sqrt{n} \\ -\sqrt{n} & 1 \end{pmatrix} \oplus (a_0 + ng - \alpha) I_n \\ & \oplus (g + n\beta + \alpha - b_0) I_n \oplus (g + n\beta) I_{n^2 - 1}. \end{aligned} \tag{24}$$

The last block is absent when $d = 2$. To see positivity, first $g = (\beta - c/d)x + cnt/d \geq 0$. Also $s(x + t) \leq x^2$, since $x, t \geq 0$. Therefore

$$a_0 - \alpha = \beta x^2 - \frac{2cn}{d^2} s(x + t) \geq (\beta - 2cn/d^2) x^2 \geq 0,$$

$$b_0 - \alpha = \beta t^2 + \frac{2c}{d^2} s(x+t) \le \beta t^2 + \frac{\beta}{n} x^2 = \frac{\beta}{n}.$$

Here $2cn/d^2 \le \beta$ follows from $c < \beta$ and $2n \le d^2$. Every block in (24) is positive semidefinite. The slack annihilates $|I\rangle\!\rangle = \sum_i |ii\rangle$, the Choi vector of the identity recovery. Its dual value thus equals the identity recovery's primal value, proving optimality. □

5 Exact instances and a relaxation obstruction

For $d = 2$, the formula is particularly simple. Define

$$k = \frac{3-2p}{3+p}, \quad h_0 = \frac{1+p}{2}, \quad w_0 = \sqrt{\frac{(3+p)(1-p)}{12}}, \quad A_2 = k(h_0 + w_0), \quad B_2 = k(h_0 - w_0).$$

If $\xi > A_2$ is the unique root there of

$$\xi^4 - 2(A_2 + B_2)\xi^3 + 4A_2B_2\xi^2 - (A_2B_2)^2 = 0,$$

then the joint fidelity is

$$F_1^{\text{joint}}(p, 2) = \frac{1}{4}\left(1 + \xi + \frac{A_2^2}{\xi - A_2} + \frac{B_2^2}{\xi - B_2}\right), \quad x_* = \frac{A_2}{\xi - A_2}, \quad z_* = -\frac{B_2}{\xi - B_2}.$$

This is also obtained directly from the circle objective $[1 + A_2(x^2 + 2x) + B_2(z^2 - 2z)]/4$. At $p = 9/13$, $A_2 = 105/208$, $B_2 = 49/208$, and $F_1^{\text{joint}} \simeq 0.6462632754$.

For a qutrit instance choose $q = 1/2$, hence $p = 8/11$. Then

$$\lambda_3 = \frac{4}{1617}, \quad A = \frac{527}{1617}, \quad C_0 = \frac{578}{4851}.$$

Writing $T = 1617\mu$, the specified quartic becomes

$$T^4 - 1054T^3 - 440147T^2 + 155349060T - 12068784500 = 0, \qquad 1305 < T < 1306. \tag{25}$$

Equations (16), (20), and (23) give exact encoder and decoder certificates over this real root field. Numerically $x_* \simeq 0.9632339434$, $z_* \simeq -0.1899741696$, and

$$F_1^{\text{joint}}(8/11, 3) = f_3(8/11) \simeq 0.8922491436 < F^{\text{unrestricted}}(8/11, 3) = \frac{69}{77}.$$

In particular, Theorem 2 proves the global certificate

$$\operatorname{tr}(K(V)X) \le f_3(8/11) \quad \text{for all } V^\dagger V = I_3,\ X \succeq 0,\ \operatorname{tr}_{\text{out}} X = I_3. \tag{26}$$

The root isolation, fixed-decoder gap matrix and recovery criterion together certify this value over the entire complex joint feasible set.

There is an exact obstruction to the simplest polynomial certificate even with the decoder fixed. No identity expressing $f_d(p) - F(V, \operatorname{tr}_S)$ as a sum of squares of affine real polynomials in V, plus constant real multipliers of the equations $V^\dagger V - I = 0$, can hold. Indeed let $C_i = QD_i^\dagger$, $i = 0, \ldots, d-1$, be the unrestricted optimal encoder's Kraus operators [1, Eq. (19)]. Assign probability $1/(2d)$ to each $V = \pm\sqrt{d}\, C_i$. This distribution is not supported on isometries, but it satisfies

$$\mathbb{E}[V^\dagger V] = I, \qquad \mathbb{E}[F(V, \operatorname{tr}_S)] = \sum_i F(C_i, \operatorname{tr}_S) = \lambda_1 > f_d(p).$$

The last strict inequality follows from the rank requirement for unrestricted optimality in Ref. [1], Sec. III. Taking expectations in the hypothetical identity is a contradiction. Thus a quadratic moment relaxation loses precisely the encoder constraint at issue. This obstruction does not rule out degree-four certificates. The qutrit proof above bypasses this relaxation by applying the recovery criterion before optimizing the encoder.

6 A uniform question beyond this family

Equation (16) of Ref. [1] already bounds the gap with both encoder and decoder optimized. This family approaches its small-error ceiling to second order as d grows; it does not attain that bound at finite d. A precise broader question is the sharp coefficient at fixed logical and physical dimensions. For qubits encoded into dimension four, let

$$c_{2,4} = \limsup_{\eta \downarrow 0} \sup_{\substack{\mathcal{N}:\mathsf{L}(\mathbb{C}^4)\to\mathsf{L}(\mathbb{C}^4)\ \mathrm{CPTP} \\ 0<\varepsilon^{\mathrm{opt}}(\mathcal{N})\le\eta}} \frac{\Delta^{\mathrm{opt}}(\mathcal{N})}{[\varepsilon^{\mathrm{opt}}(\mathcal{N})]^2},$$

where $\varepsilon^{\mathrm{opt}} = 1 - F^{\mathrm{unrestricted}}$ and $\Delta^{\mathrm{opt}} = F^{\mathrm{unrestricted}} - F_1^{\mathrm{joint}}$ at $\tau = I_2/2$. Their Eqs. (16) and (25) give

$$\frac{1}{9} \le c_{2,4} \le \frac{1}{4}.$$

Resolving this interval requires either a new noise family approaching $1/4$ or a quantitative incompatibility among the near-equality conditions in the rank-one replacement bound and joint optimality. This is distinct from finding an exact solution for the present model.

Verification and computer assistance. The conclusions above are mathematical proofs, not Lean formalizations. The accompanying source checks symbolic identities and finite exact matrix certificates with rational arithmetic; numerical approximations serve to identify values. Haobo Ma and Wenlin Zhang used AI tools for search, double-checking and proofreading.

References

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